Longitudinal Data Analysis

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1 Introduction

Chronic kidney disease is a significant public health problem and one of the leading causes of death associated with non-communicable diseases. In 2017, approximately 1.2 million deaths were attributed to CKD, with a global prevalence of 9.1%, reflecting a 29.3% increase since 1990. End-Stage Kidney Disease (ESKD) is the most severe form of CKD, characterized by a substantial decline in kidney function that requires interventions such as hemodialysis or kidney transplantation to sustain life.

Patients with ESKD frequently experience complications, including anemia, primarily due to the kidneys' reduced ability to produce erythropoietin (EPO). EPO is a crucial hormone that stimulates red blood cell production (erythropoiesis) in the bone marrow, and its deficiency leads to reduced hemoglobin levels. Additionally, hemodialysis can further reduce iron stores, exacerbating anemia. In severe cases, anemia can result in cardiovascular complications, significantly affecting the overall health and quality of life of these patients.

To effectively manage anemia in patients with ESKD, a comprehensive approach is essential. The current strategy primarily includes the administration of recombinant EPO. Since adequate iron levels are necessary for erythropoiesis, it is crucial to evaluate and maintain sufficient iron stores. Accordingly, previous studies have shown that iron deficiency can reduce the effectiveness of EPO, leading to persistent anemia despite treatment. Nevertheless, some patients do not respond to EPO therapy even when adequate iron stores are present. Therefore, this study aims to investigate how changes in hemoglobin levels over time are influenced by EPO dosing, iron status, and demographic factors, providing insights to improve anemia management in patients with ESKD receiving hemodialysis.

1.1 Objective

The main objective of the present study is to investigate how hemoglobin levels change over time in patients with renal deficiency receiving hemodialysis and analyze the relationship between the dose of EPO administered. Addtionally, we will try to assess the effect of iron deficiency status as well as the effect of demographic factors (age and sex) on these changes.

2 Materials and Methods

2.1 Data Description

The dataset used for this study includes monthly measurements of hemoglobin (Hb, g/dl, continuous), EPO dose (IU/kg/week, IU: International Unit, discrete), and iron deficiency status (categorical, indicated when

serum ferritin levels are below 100 μg and/or transferrin saturation is below 20%). Additionally, demographic data, including sex (categorical) and age (discrete), were collected.

The dataset contained data from 3823 patients receiving hemodialysis. The total follow-up period lasted 6 months. However, not all patients attended every visit and a large number were lost to follow-up after the 4th month, resulting in an unbalanced dataset. On average patients attended approximately 2 to 3 visits. Table 1 summarizes the number of patients according to their visit counts.

During the follow-up, patients received EPO treatment to maintain hemoglobin levels within the desired range, with hemoglobin measurements used to adjust the EPO dose for the following month. Given that there is no immediate biological effect of EPO on hemoglobin, and the doses in the dataset are based on current hemoglobin levels, further analysis might reflect clinicians' decisions rather than the true effect of EPO on hemoglobin over time. Therefore, to capture the effect of EPO administration on changes in hemoglobin levels over time, we generated two variables: lagged dose and lagged Hb.

Lagged dose represents the EPO dose received in the previous month. Since the effect of EPO on hemoglobin levels typically becomes evident between 2 to 6 weeks after administration, this variable accounts for the delayed effect of EPO on hemoglobin levels, allowing for a more accurate analysis of the relationship between past EPO doses and subsequent changes in hemoglobin. Lagged hb represents the hemoglobin level from the previous month. As hemoglobin levels tend to remain relatively stable over time, with gradual changes, this variable accounts for the temporal persistence of hemoglobin levels, allowing for a better understanding of the natural progression of hemoglobin levels over time.

Table 1: Number of Patients by Visit Count

Visits	1	2	3	4	5	6
Number of Patients	535	828	1180	780	127	28

2.2 Descriptive Analysis

To understand the data structure and guide model selection, we examined the mean structure, variance structure, and covariance structure of the dataset. The mean structure was assessed by analyzing the average evolution of hemoglobin levels (g/dl) over six months for a random sample of subjects. Additionally, we examined trends in hemoglobin levels over time for key variables, including dose EPO, iron status, age, and sex. Although the time points in this study are fixed, the number of repeated measurements varies across subjects, making the data unbalanced. To capture the overall trend and evaluate the need for a polynomial term, LOESS smoothing was applied. The variance structure was assessed for heteroscedasticity by examining mean squared residuals over time. The variance was found to be non-constant, limiting the suitability of certain analyses, such as the semivariogram. Given the non-constant variance, we further investigated the covariance structure by plotting correlation structure from observed data.

2.3 Summary Statistics

Given that this is a prospective study with repeated measurements, classical methods cannot be employed to assess the effect of EPO dose on hemoglobin over time. In this scenario, alternative techniques that summarize the number of measurements can be employed. The Area Under the Curve (AUC) for each individual and the analysis of increments in hemoglobin levels per were considered for this purpose because of their ability to handle unbalanced data and are not affected by multiple testing.

The AUC captures the overall change in hemoglobin levels throughout the treatment period, allowing for straightforward comparisons across patients. However, it does not provide information about how hemoglobin levels change over time, and assumes a relatively stable response of EPO. On the other hand, increment in hemoglobin levels was calculated as the difference between the first and last measurements, offering insights into the net effect of EPO treatment over time. While this approach simplifies the assessment of treatment impact, it is sensitive to gaps in data, which were observed in our dataset. Therefore, the results of these analyses should be interpreted with caution.

As both methods summarize the overall change in hemoglobin over the follow-up period, they inherently capture the treatment effect based on the baseline EPO dose. Therefore, there was no need to include lagged dose and lagged hemoglobin in their analyses.

2.4 Modeling

2.4.1 Multivariate Regression Model

A general multivariate linear regression model was initially considered for our analysis. This preliminary model is specified as:

$$Y_i = X_i\beta + \epsilon_i$$

where $Y_i = (Y_{i1}, \ldots, Y_{in})$ is the vector of n repeated measurements for the i-th subject, X_i is the matrix of covariates,β is the vector of regression parameters and ϵ_i is a vector of residual error terms. ϵ_i is assumed to follow $\epsilon_i \sim N(0, \Sigma)$, based on a covaraince matrix Σ that accounts for the correlation within each subject's repeated measurements.
For this model, we included lagged dose and lagged hemoglobin to capture the time-dependent nature of the data and the delayed effects of EPO on hemoglobin levels. Current dose was excluded due to reverse causality, as hemoglobin levels influence dosing decisions, and its inclusion had no direct clinical relevance.

To understand the evolution of our response variable over time, we explored multiple mean and covariance structures. The most parsimonious mean structure was identified by sequentially fitting different combinations of covariates, starting with a full model including interaction terms and progressively reducing the complexity. After the mean structure was determined, several covariances structures were tested to model the within-subject variability balancing goodness-of-fit and model simplicity.

All comparisons between models were based on Likelihood Ratio Tests (LRTs), under the hypothesis that the simpler model with fewer parameters would provide an adequate fit to the data compared to the more complex model. Inferences were made using classical Maximum Likelihood estimation on the final selected model, adjusting the standard errors to make them robust against potential misspecifications.

2.4.2 Two-Stage Model

Before conducting a two-stage analysis, we first assessed how well the observed longitudinal profiles could be captured by a specified linear regression model. To do this, we evaluated the coefficient of determination, R^2, which quantifies the proportion of total variability in the response variable that can be accounted for by the covariates under consideration. The value of R^2 ranges between 0 and 1, where values closer to 1 indicate a stronger fit of the model to the data.

For each subject, we computed an individual R_i^2 to determine how well the linear regression model described their longitudinal data.

To provide an overall assessment of the model's goodness of fit across all subjects, we used a meta-analytic measure of R^2, defined as:

$$R^2_{\text{meta}} = \frac{\sum_{i=1}^{N}(\text{SSTO}_i - \text{SSE}_i)}{\sum_{i=1}^{N}\text{SSTO}_i}$$

where SSTO_i represents the total sum of squares for subject i, and SSE_i denotes the error sum of squares for subject i. This aggregate measure reflects the proportion of total within-subject variability that can be explained by the linear regression models from the first stage of the analysis.

2.4.3 Stage 1

In Stage 1, we fitted subject-specific regression models using hemoglobin as a linear function of time and time-varying covariates, as well as with higher-order polynomial time effects to explore potential improvements in model fit. Specifically, we considered linear, quadratic, and cubic time effects as a preliminary analysis. To assess whether extending the model to include higher-order time effects was warranted, we used the F_{meta} statistic.

The F_{meta} statistic served us as a tool for comparing models in order to determine if the inclusion of additional parameters significantly improves model fit. When a comparison of the linear time effect model to the quadratic time effect model was carried on, the F_{meta} statistic indicated a significant improvement. When extending from the quadratic model to the cubic model, the F_{meta} statistic yielded a significant value as well. Although higher-degree time effects resulted in increased R^2 values, 0.86 for the linear model, 0.90 for the quadratic model, and 0.98 for the cubic model—each additional degree of complexity came with a trade off in interpretability.
To achieve a more interpretable model, we opted to use the linear time effect model, despite a slight loss in explanatory power.

For each subject i ($i = 1, 2, \ldots, N$), we observe Hb measurements at times t_{ij} ($j = 2, 3, \ldots, n_i$), where n_i is the number of observations for subject i after the baseline measurement.
We model:

$$Y_{ij} = \beta_{i0} + \beta_{i1} \times \text{Month}_{ij} + \beta_{i2} \times \text{Iron}_{ij} + \beta_{i3} \times \text{Dose}_{i,j-1} + \beta_{i4} \times Y_{i,j-1} + \epsilon_{ij}$$

Where:

- Y_{ij}: Hemoglobin (Hb) level for subject i at time t_{ij}.
- Month_{ij}: Current month for subject i.
- Iron_{ij}: Iron status for subject i at time t_{ij}.
- $\text{Dose}_{i,j-1}$: EPO dose for subject i at previous time $t_{i,j-1}$.
- $Y_{i,j-1}$: Hb level for subject i at previous time $t_{i,j-1}$.
- β_{i0}: Subject-specific intercept.
- $\beta_{i1}, \beta_{i2}, \beta_{i3}, \beta_{i4}$: Subject-specific coefficients.
- ϵ_{ij}: Error term, $\epsilon_{ij} \sim N(0, \sigma^2)$.

2.4.4 Stage 2

For each coefficient k ($k = 0, 1, 2, 3, 4$):

$$\beta_{ik} = \gamma_{0k} + \gamma_{1k} \times \text{Age}_i + \gamma_{2k} \times \text{Sex}_i + b_{ik}$$

Where:

- β_{ik}: Estimated coefficient k for subject i from Stage 1.
- γ_{0k}: Fixed intercept term for coefficient k.
- γ_{1k}, γ_{2k}: Fixed effect coefficients for Age and Sex on β_{ik}.
- Age_i: Age of subject i at study entry.
- Sex_i: Sex of subject i
- b_{ik}: Random effect for subject i and coefficient k, $b_{ik} \sim N(0, \tau_k^2)$.

2.4.5 Linear Mixed Effect Model

The linear mixed model was selected to capture both the fixed effects and the random variation between individuals over time. The model is specified as follows:

$$Y_i = X_i\beta + Z_i b_i + \varepsilon_i \tag{1}$$

where Y_i is the vector of repeated hemoglobin (Hb) measurements for the i-th subject, X_i is the matrix of covariates, β is the vector of fixed-effect parameters, Z_i is the design matrix for the random effects, b_i represents the random intercepts for each subject, and ϵ_i is the residual error term. The random intercepts b_i are assumed to follow a normal distribution $b_i \sim N(0, D)$, where D is the variance-covariance matrix for the random effects. The residuals ϵ_i are also assumed to follow a normal distribution $\epsilon_i \sim N(0, \Sigma_i)$, where Σ_i models the correlation structure within each subject over time.

To test for the presence of serial correlation, we compared models with only measurement error to those with exponential and Gaussian serial correlation structures. Fit statistics, calculated via restricted maximum likelihood (REML), were used to determine the best-fitting correlation structure. A likelihood ratio test (LRT) was conducted to compare models, with the Gaussian serial correlation structure ultimately selected based on improvements in model fit.

The structure of random effects was examined by testing whether adding a random slope for time would significantly improve the model fit compared to a simpler model with only a random intercept. The result was evaluated based on the asymptotic null distribution, where$-2\ln\lambda$ follows a $\chi^2_{1:2}$ distribution, under the null hypothesis that the random intercept-only model is sufficient.

3 Results

3.1 Exploratory Data Analysis

3.1.1 Individual Profile Plots

As shown in Figure 1, for better visualization of the patterns in the data, we randomly sampled one percent of individuals in the population, in which each line represents an individual's hemoglobin levels over time. This plot shows that individuals entered the study with varying baseline hemoglobin levels. This initial high variability between subjects suggests the need for a random intercept in the models, however, we cannot conclude if variability between subjects differs significantly over time, further statistical testing is needed to assess this. —

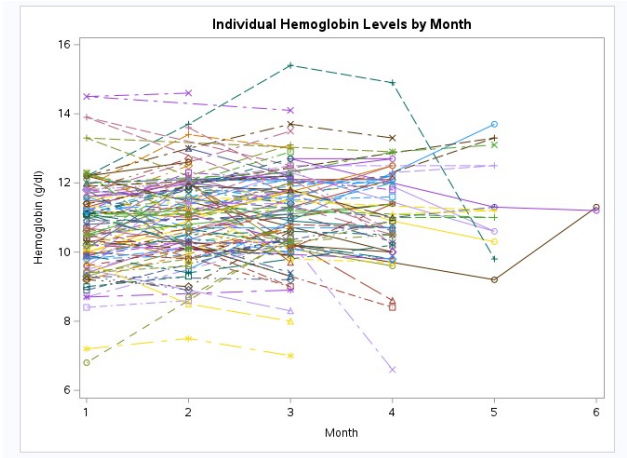

Figure 1: Individual hemoglobin levels by month.

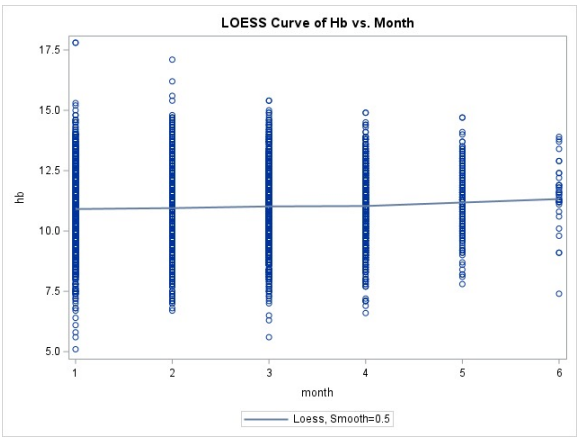

Figure 2: LOESS curve of Hb over months

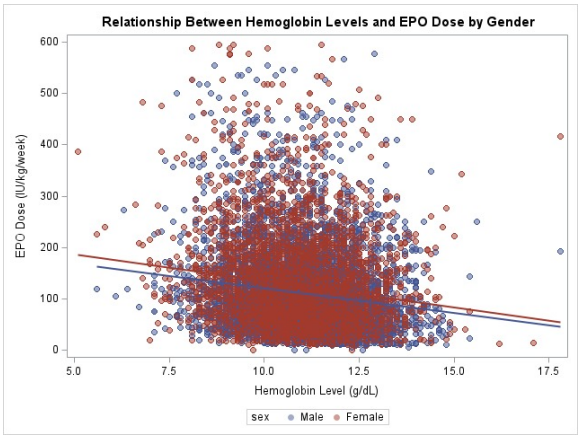

Figure 3: Hemoglobin Levels and EPO Dose by Gender

3.1.2 Mean Structure

Upon examining figure 2, it is evident that the LOESS smoothing curve for the mean structure exhibits a relatively flat trend over time. This suggests that the relationship between time and hemoglobin levels is likely linear, with no strong evidence to support the need for a polynomial transformation. Nevertheless, we could include a quadratic term during the model-building process as a precaution.

Figure 3 shows the relationship between Hemoglobin Levels (g/dL) and EPO Dose (IU/kg/week) for Male and Female, illustrating a mild inverse relationship between hemoglobin levels and EPO dosage, that indicates individuals with higher hemoglobin levels generally require lower doses of EPO. There may also be a slight gender difference in this relationship, with one gender potentially needing higher doses at comparable hemoglobin levels. However, this difference seems small and requires further investigation to confirm.

Figure 4 shows hemoglobin changes across months for different age groups. While all groups experienced increases by the sixth month, subjects under 30 began with the lowest levels and surpassed those over 60 by the end. The 30-60 age group maintained the highest levels throughout most of the study. This suggests a potential effect of age on hemoglobin, warranting further exploration. The sharp increase in the under-30 group in the last month may be worth investigating to understand younger individuals' responses to dose adjustments.

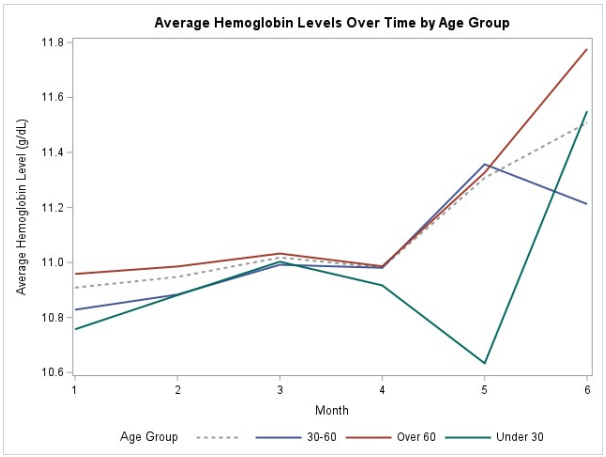

Figure 4: HB Levels Across Months by Age

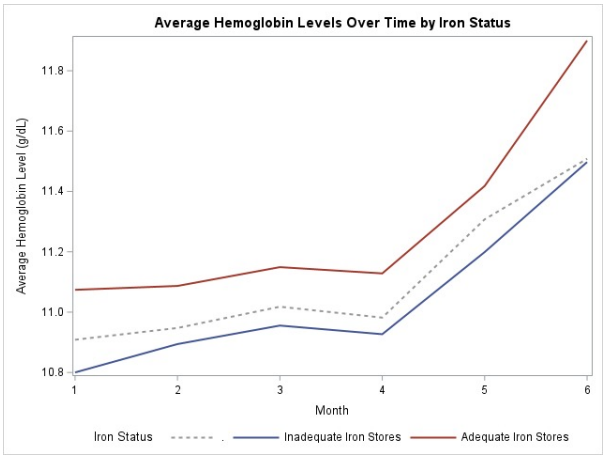

Figure 5: HB Levels Across Months by Iron

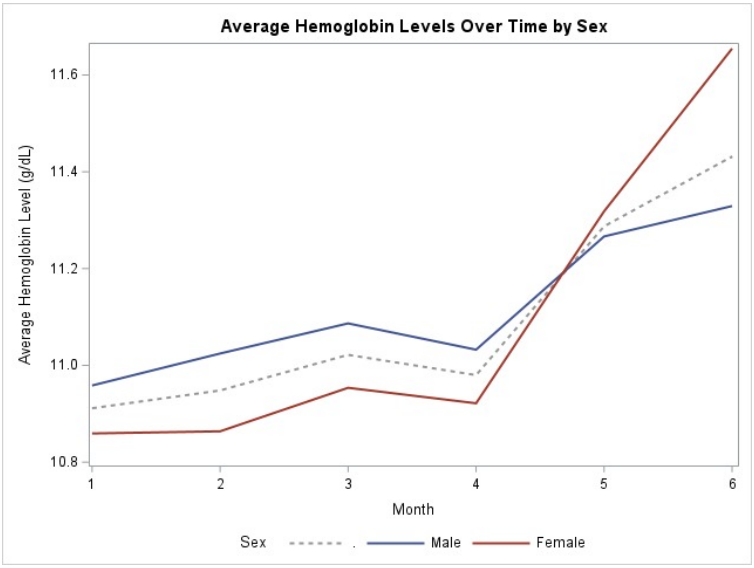

Figure 6: HB Levels Across Months by Sex

The Figure 5 shows that adequate iron stores improve hemoglobin production over time, particularly in later months, compared to those with insufficient iron, who show slower growth. In contrast, those with insufficient iron stores tend to have lower baseline hemoglobin levels and slower growth.

The figure 6shows that males generally started with slightly higher hemoglobin levels than females. However, females experienced a sharper rise in the final two months, resulting in higher average levels by the study's end. This trend may indicate differences in response to treatment or external factors affecting hemoglobin levels, which could be further explored.

3.1.3 Variance Structure

As illustrated in Figure 7, the unsmoothed plot of Mean Squared Residuals (MSR) after adding lagged version of dose and hb and accounting our variance from second month to sixth reveals a clear decline in variability across time, aligned with slight fluctuations. The trend is further clarified in Figure 8, where the smoothed

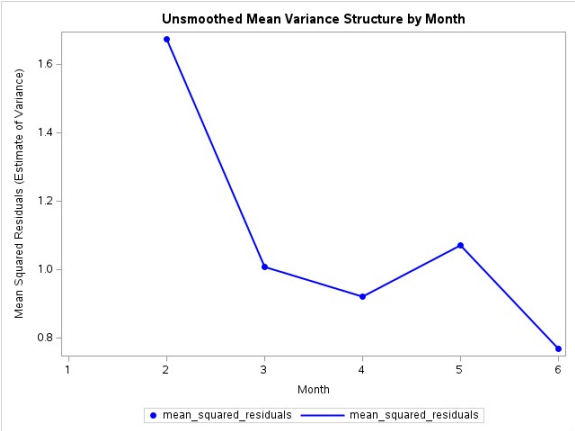

Figure 7: Unsmoothed MSR

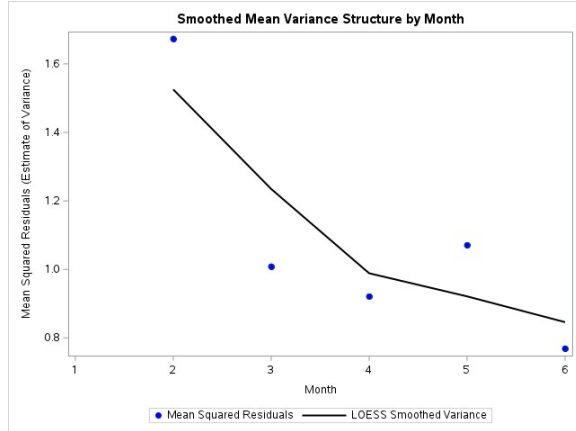

Figure 8: smoothed MSR

MSR plot highlights a consistent decrease in variability. This observed reduction in MSR indicates that the variability of observations within subjects is time-dependent and diminishes as the study progresses. This pattern of decreasing variability strongly suggests the presence of time-dependent heteroscedasticity. Such behavior aligns well with correlation structures like the Heterogeneous Toeplitz (HTOEP), which can model both the declining variances and the evolving correlations among repeated measures over time. Furthermore, the presence of heteroscedasticity in the data implies that traditional tools like the semivariogram plot are not appropriate for further analysis, as they assume constant variance across time points. This highlights the need for covariance structures that explicitly accommodate changing variances and correlations in the statistical model.

3.1.4 Correlation Structure

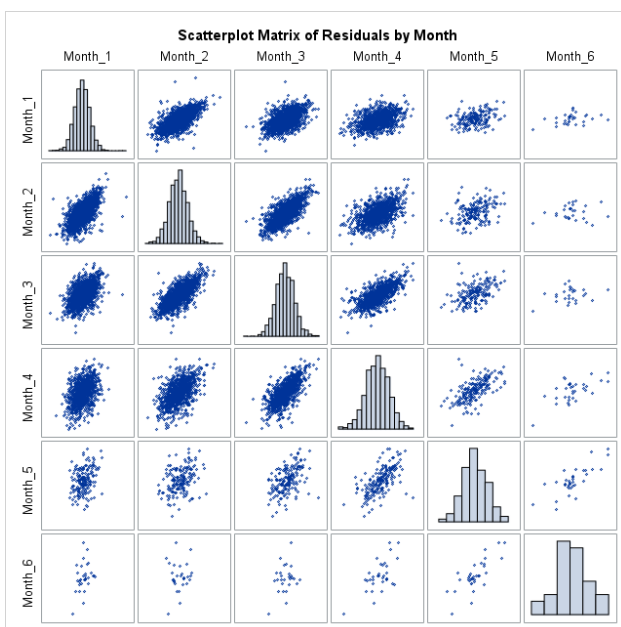

Figure 9: Scatterplot Matrix of Residual Correlations Over Time

In the figure 9, the correlation matrix of residuals plotted over months revealing an inconsistent but generally decreasing correlation as the time lag between measurements increases, suggesting that Hb measurements

become more stable toward the end of the study. The observed pattern of decaying and varying correlations, with stronger correlations between adjacent months and weaker correlations over greater time separations, indicates the need to test various correlation structures in subsequent modeling stages. Specifically, potential candidates include Heterogeneous Toeplitz (TOEPH), and Heterogeneous Autoregressive of Order 1 (ARH[1]) correlation structures.

3.2 Summary Statistics

We analyzed the results from the calculation of the time standardized AUC and increments using a multiple linear regression, adjusting for the covariates sex, age, iron and deficiency. The results of these analyses are summarized in Table 2. In the AUC model, which accounts for time by dividing cumulative AUC by each individual's total observation period, we observed a significant negative effect of the EPO dose ($\beta = -0.0029$, $p < 0.0001$) and iron deficiency ($\beta = -0.3554$, $p < 0.0001$) on hemoglobin levels. In contrast, the increments model did not reveal any significant effects of the covariates on hemoglobin levels.

However, these results should be interpreted with caution as both models exhibited issues with singularity of the X'X matrix, leading to the use of a generalized inverse by SAS to solve the normal equations. This suggests potential collinearity or insufficient variability in the predictors, which may affect the reliability of the results.

Given we did not observe collinearity between the variables included in the models, we concluded that summarizing the hemoglobin levels using these techniques likely resulted in a loss of crucial information needed to capture the change in hemoglobin levels over time. Moreover, since the majority of patients had only 2 or 3 measurements taken during the study period, the limited variability in the outcome measures (standardized AUC and Hb increment) reduced the model's ability to estimate the true effects of the covariates. Therefore, further analysis using models that account for repeated measures and temporal variation in hemoglobin levels were performed.

Table 2: Analysis of the Summary Statistics

Parameter	**Estimate**	**Standard Error**	**t Value**	**p-value**
AUC Standardized Model				
Intercept	11.5684	0.1548	74.74	< 0.0001
Epo dose	-0.0029	0.0004	-8.06	< 0.0001
Sex (Female)	-0.1412	0.0644	-2.19	0.0286
Iron deficiency	-0.3554	0.0699	-5.09	< 0.0001
Age	-0.0018	0.0022	-0.86	0.3921
Increment Model				
Intercept	0.2174	0.0946	2.30	0.0216
Epo dose	-0.0007	0.0002	-0.28	0.7763
Sex (Female)	0.0479	0.0399	1.20	0.2309
Iron deficiency	0.0716	0.0423	1.69	0.0907
Age	-0.0035	0.0013	-2.65	0.0081

3.3 Multivariate Regression

A preliminary mean structure was defined by including main effects, interaction terms, and quadratic effects. Based on the evolution of hemoglobin (Hb) levels over time, we initially fit a fully saturated model with an unstructured covariance structure.

Table 3: Model Reduction and Covariance Structure Selection Results

Model	**Mean**	**Cov**	**Parameters**	**-2 LogLik**	**Test**	χ^2	**df**	**p**
1	Unstr.	Unstr.	39	17803.3	-	-	-	-
2	No Month^2 * Age	Unstr.	38	17803.3	(1 vs 2)	0	1	1
3	No Month^2 * Lagged Dose	Unstr.	37	17803.7	(2 vs 3)	0.4	1	0.52709
4	No Month^2 * Sex	Unstr.	35	17804.2	(3 vs 4)	0.5	2	0.77880
5	No Month^2 * Iron	Unstr.	33	17804.3	(4 vs 5)	0.1	2	0.95123
6	No Month^2 * lagged Hb	Unstr.	32	17807.2	(5 vs 6)	2.9	1	<0.08860
7	No Month^2	Unstr.	30	17810.1	(5 vs 7)	11.6	2	<0.00889
8	No Month * Age	Unstr.	32	17805.5	(5 vs 8)	2.4	1	0.12138
9	No Month * Sex	Unstr.	30	17807.3	(8 vs 9)	3.6	2	0.16530
10	No Month * Iron	Unstr.	28	17808.0	(9 vs 10)	1.4	2	0.49659
11	No Month^2	Unstr.	27	17812.3	(10 vs 11)	8.6	2	0.00336
12	Model 10	ARH(1)	19	17830.4	(10 vs 12)	9	22.4	0.00769
13	Model 10	TOEPH	22	17816.0	(10 vs 13)	6	8	0.2381

Table 3 shows the model reduction by LRTs until the most parsimonious mean structure was found. Our final model included the main effects for Month, Age, Sex, Iron, lagged dose, lagged hb and the interaction terms between month and month, month and lagged dose, month and lagged hb and the interaction term between quadtradic time and Lagged Hb. Given the exploratory analysis indications that variance was not constant and the correlation between measurements decayed with greater time intervals, we tested Heterogeneous Auto-regressive and Heterogeneous Toeplitz covariance structures. Based on the LRT results, Heterogeneous Toeplitz was found to be the most parsimonious covariance structure as a simpler alternative to the unstructured but not introducing a significant drop in fit quality, which confirmed the observed patterns during exploration. The final multivariate model was defined as follows:

$$\begin{aligned}\text{Hb}_{ij} = \beta_0 &+ \beta_1\text{Month}_j + \beta_2\text{Age}_i + \beta_3\text{Sex}_i + \beta_4\text{Iron}_i \\ &+ \beta_5\text{Lag_Dose}_{ij} + \beta_6(\text{Month}_j \times \text{Age}_i) \\ &+ \beta_7(\text{Month}_j \times \text{Dose}_{ij}) + \beta_8(\text{Month}_j \times \text{Iron}_i) + \epsilon_{ij}\end{aligned}$$

The parameter estimates for the final model are presented in Table 4, as well as their naive and standard errors and p-values. At the 5% level of significance, it can be concluded that Hb levels are not changed over time based on age, dose or iron deficiency status. The only conclusions that can be made are that prior Hb values influence Hb levels at later time points ($p < 0.0001$), in a non-linear pattern over time (-0.0456, $p = 0.0001$).

Table 4: Solution for Fixed Effects

Effect	**Estimate**	**SE**	**Robust SE**	**DF**	**t**	**p > |t|**
Intercept	14.1314	1.1348	1.2555	2902	11.26	< 0.0001
Month	-3.3596	0.7494	0.8412	3155	-3.99	< 0.0001
Age	0.0012	0.0014	0.0014	2902	0.87	0.3829
Sex (Male)	0.1135	0.0411	0.0411	2902	2.77	0.0057
Iron (adequate)	0.1014	0.0304	0.0310	616	3.27	0.0011
Lagged Hb	-0.3007	0.1014	0.1112	3155	-2.7	0.0069
Lagged Dose	0.0008	0.0005	0.0005	3155	1.55	0.1214
Month × Month	0.5189	0.1182	0.1341	3155	3.87	0.0001
Month × Lagged dose	0.0002	0.0017	0.0750	3155	1.15	0.2516
Month × Lagged Hb	0.2953	0.0671	0.0750	3155	3.94	< 0.0001
Month × Month × Lagged Hb	-0.0456	0.0105	0.0120	3155	-3.81	0.0001

3.4 Two-Stage Analysis

3.5 Assessing Goodness of fit

It would be valuable to assess how well the specified linear regression model captures individual longitudinal profiles. Calculating the coefficient of determination R^2 for each subject provided us with insights into the model's fit at the individual level. Aggregating these individual R_i^2 values into an overall meta-analytic R^2_{meta} offered a comprehensive measure of the model's goodness-of-fit across all subjects. The linear model had an already high R^2 of 0.83, meaning that within subject variability across all subjects already explains a large proportion of the total variance.

Table 5: Meta-Analytic R^2 Values for models with linear, quadratic and cubic time effect

Metric	Value
$R^2_{\text{meta_M1}}$	0.86
$R^2_{\text{meta_M2}}$	0.90
$R^2_{\text{meta_M3}}$	0.98

In order to compare our models and to validate the necessity of the extensions, we calculated the F_{meta} which involved the difference in the sum of squared errors between the models for each subject, divided by the difference in the number of parameters, and normalizing by the SSE of the more complex model over the total degrees of freedom. $\text{SSE}_{i,M1}$ and $\text{SSE}_{i,M2}$ represent the sum of squared errors for subject i under Model 1 and Model 2, respectively. The numerator degrees of freedom are calculated as $d_{M2} - d_{M1}$, and the denominator degrees of freedom df_{den} are given by $n_{\text{total}} - d_{M2}$.

$$F_{\text{meta}} = \frac{\sum_{i=1}^{N}(\text{SSE}_{i,M1} - \text{SSE}_{i,M2})/(d_{M2} - d_{M1})}{\sum_{i=1}^{N}\text{SSE}_{i,M2}/(n_{\text{total}} - d_{M2})}$$

The same process was carried on for Model2 and Model3.
As mentioned already, the inclusion of a quadratic time effect increased the coefficient of determination significantly to 0.90, while the cubic effect increased it even further to 0.98. The Fmeta statistics proved to be significant, however we opted to proceed with a simple and linear model, considering that it already adequately explains the data without compromising interpretability. Furthermore, through visual analysis it was observed that although the R meta was improving, that was due to the inherent unbalanced nature of our data. Subjects with more observations were benefiting more the inclusion of non linear effects, while subjects fewer measurements were not.

3.5.1 Stage 1

Our objective in stage 1 was to estimate subject-specific regression coefficients by fitting a linear regression model for each subject separately, using the time - Month and time-varying covariates within subject, Iron and Dose.

Table 8 presents the summary statistics for the subject-specific intercepts and slopes. The average subject-specific intercept was estimated to be 10.58 with a sd of 33.52 and the average subjects specific slopes for Month, Iron and Dose were 0.059,, 0.063 and -0.001 respectively. We then proceeded by making use of stage's 1 estimates in stage's 2 analysis.

Table 6: Parameter Estimates

ID	Intercept	Month	Iron	Lagged Dose	Lagged Hb
100001	5.79	-1.412	0.0000	0.00000	0.8750
100002	-10.50	-0.400	0.0000	0.00000	2.0000
100003	12.79	0.000	0.0000	-0.00889	0.0000
100004	16.30	0.000	0.0000	0.00000	-0.5000
...	...	...	...	...	...
132457	12.30	-0.300	0.9000	0.00000	0.0000
132458	11.82	-0.250	0.0000	0.00000	0.0000
132459	11.05	1.294	0.0000	0.00000	-0.3529
132460	11.80	-0.400	-0.2000	0.00000	0.0000

Table 7: The MEANS Procedure

Variable	Label	N	Mean	Std Dev	Minimum	Maximum
Intercept		2905	10.3103	47.3437	-1968.43	692.3545
Slope_Month		2905	0.0485	3.5179	-43.22	175.7
Coef_Iron		2905	0.0295	1.7265	-6.2	86.9
Coef_LaggedDose		2905	0.0020	0.2766	-5.0	8.967
Coef_Laggedhb		2905	-0.0193	1.2017	-39.9091	18.6667

3.5.2 Stage 2

Our primary goal in this analysis was to obtain an initial understanding of the trends in hemoglobin levels over time and to assess the effect of age on these trends among patients. After having estimated the subject specific coefficients, we xamined their variability by regressing them on the time-invariant covariates to understand their influence on the hb trajectories.

Table 8: Parameter Estimates for Intercept

Variable	Label	DF	Parameter Estimate	Standard Error	t Value	Pr
Intercept	Intercept	1	5.53267	4.64366	1.19	0.2336
Age	age	1	0.03924	0.06051	0.65	0.5167
Sex	sex	1	1.62395	1.81163	0.90	0.3701

Table 9: Parameter Estimates for Slope_Month

Variable	Label	DF	Parameter Estimate	Standard Error	t Value	Pr
Intercept	Intercept	1	0.74147	0.34665	2.14	0.0325
Age	age	1	-0.00859	0.00452	-1.90	0.0572
Sex	sex	1	-0.10514	0.13524	-0.78	0.4370

Table 10: Parameter Estimates for Coef_Iron

Variable	Label	DF	Parameter Estimate	Standard Error	t Value	Pr
Intercept	Intercept	1	0.13970	0.17027	0.82	0.4120
Age	age	1	-0.00107	0.00222	-0.48	0.6285
Sex	sex	1	-0.02922	0.06643	-0.44	0.6601

Table 11: Parameter Estimates for Coef_Dose

Variable	Label	DF	Parameter Estimate	Standard Error	t Value	Pr
Intercept	Intercept	1	0.01722	0.02702	0.64	0.5239
Age	age	1	-0.00004626	0.00035210	-0.13	0.8955
Sex	sex	1	-0.00865	0.01054	-0.82	0.4117

Table 12: Parameter Estimates for Coef_hb

Variable	Label	DF	Parameter Estimate	Standard Error	t Value	Pr
Intercept	Intercept	1	-0.03422	0.11857	-0.29	0.7729
Age	age	1	0.00015553	0.00155	0.10	0.9198
Sex	sex	1	0.00386	0.04626	0.08	0.9335

In summary, the two-stage analysis indicates that age does not have a significant effect on the baseline hemoglobin levels, the rate of change in Hb over time, or the influence of iron status and EPO dose on Hb levels among patients undergoing hemodialysis. The R-squared values for all models were extremely low (ranging from 0.0005 to 0.0012), suggesting that age and sex explain a negligible proportion of the variability in the subject-specific coefficients derived from the first stage.

3.6 Linear Mixed Model

To investigate the evolution of hemoglobin (Hb) levels in hemodialysis patients over time, a linear mixed model was fitted, with an initial mean structure selected based on previous analysis, including linear and quadratic time effects, age, sex, iron deficiency status, and lagged values for both EPO dose and Hb levels to account for potential delayed responses, as well as their interactions.

We examined the need for serial correlation by fitting models with only measurement error, as well as models with exponential and Gaussian serial correlation structures. Fit statistics based on REML, shown in Table [x], suggest that including a Gaussian serial correlation structure improves model fit.

To determine the best random effects structure, we tested whether adding a random slope for time improved the model fit compared to a simpler model with only a random intercept. The p-value obtained was $p = 0.5897$, indicating no significant improvement with the addition of a random slope. Therefore, the random intercept-only model was retained as it sufficiently captured the random effects structure for time.

After the mean structure reduction the final linear mixed model was defined as follows:

$$\begin{aligned}
\text{Hb}_{it} = {} & \beta_0 + \beta_1 \text{Age}_i + \beta_2 \text{Sex}_i + \beta_3 \text{Month}_{it} + \beta_4 \text{Month}_{it}^2 \\
& + \beta_5 \text{Iron}_{it} + \beta_6 \text{LagDose}_{it} + \beta_7 \text{LagHb}_{it} \\
& + \beta_8 (\text{Month}_{it} \times \text{Age}_i) + \beta_9 (\text{Month}_{it} \times \text{Iron}_{it}) \\
& + \beta_{10} (\text{Month}_{it}^2 \times \text{Iron}_{it}) + \beta_{11} (\text{Month}_{it} \times \text{LagDose}_{it}) \\
& + b_{0i} + \epsilon_{it}
\end{aligned}$$

where βs are the fixed effects, b_{0i} is the random intercept for each patient, and ϵ_{it} is the residual error term.

Table 13: Parameter Estimates for the Final Linear Mixed Effects Model

Effect	Estimate	Standard Error	t-value	p-value
Intercept	9.5483	0.2315	41.25	< 0.0001
Month	0.2064	0.0859	2.40	0.0164
Month^2	-0.01305	0.01115	-1.17	0.2423
Age	0.006071	0.002583	2.35	0.0188
Sex	-0.1361	0.04381	-3.11	0.0019
Iron	0.5725	0.1464	3.91	< 0.0001
LagDose	-0.00013	0.000415	-0.32	0.7518
LagHb	0.07787	0.009986	7.80	< 0.0001
Month $\times$ Age	-0.00186	0.000836	-2.22	0.0263
Month $\times$ Iron	-0.2971	0.1089	-2.73	0.0064
$\text{Month}^2 \times$ Iron	0.04211	0.01884	2.19	0.0288
Month $\times$ LagDose	0.000335	0.000141	2.38	0.0174

Table 13 shows the parameter estimates for the final linear mixed effects model. Older patients have a slower increase in Hb over time (-0.00186, $p = 0.0263$), suggesting age-related dosing adjustments may be beneficial. Iron-deficient patients exhibit a stronger and non-linear Hb response over time compared to those with adequate iron (0.2971, $p = 0.0064$), indicating a need for targeted iron supplementation. Additionally, the effect of past EPO dose on Hb levels varies over time (0.000335, $p = 0.0174$), implying that EPO dosing strategies should be flexible and adapted regularly to optimize treatment.

Table 14: Covariance Estimates for the Linear Mixed Model

Covariance Parameter	Estimate
UN(1,1) - Random Intercept Variance	0.5998
Variance	0.7333
SP(GAU) - Serial Correlation	1.9423
Residual Variance - Measurement Error	0.2892

The covariance estimates are found in table 14 and show that the variability of Hb over time is mostly explained by the Gaussian serial correlation (1.9423), followed by the random intercept (0.5998) and the masurement error (0.2892).

3.6.1 Comparison Between Linear Mixed Model and Multivariate Model

In the linear mixed effects model, factors such as age, iron deficiency, and past EPO dose showed significant effects on hemoglobin (Hb) levels over time, accounting for within-subject variability and serial correlation. In contrast, the multivariate model captures correlations between repeated measures but does not explicitly model the time-based correlation or account for individual-level trends over time. This is a clear limitation for data of this nature and is reflected by the fact that the time dependent effects of past EPO dose and iron status were not even present in the most parsimonious model.

3.6.2 Comparison Between Linear Mixed Model and Two-Stage Model

In the linear mixed effects model, age, iron deficiency, and past EPO dose each showed significant effects on hemoglobin (Hb) levels over time, with older patients experiencing a slower Hb increase, iron-deficient patients exhibiting a stronger response, and a time-varying impact of past EPO doses. In contrast, the two-stage analysis did not find a significant effect of the population level variables on Hb levels over time, with R-squared values indicating that age and sex explained only a negligible portion of the variability derived from the time-varying coefficients.

This highlights a key limitation of the two-stage approach: by separating the estimation process into two stages, it loses essential information by not accounting for the within-subject correlation structure and individual trends that are captured in the mixed model. The linear mixed model offers a more accurate view of how Hb levels evolve by incorporating serial correlation and random intercepts, making it more informative for individualized treatment adjustments in a clinical setting.

4 Conclusion

We employed multiple statistical approaches for the analysis of this dataset, to understand how hemoglobin changes over time given iron status, age, sex and EPO dosage. Initially, exploratory analysis highlighted potential for non-linear trends and time-dependent variability in Hb levels, which guides our decisions in the specification of models for further analysis.

The multivariate model captured a preliminary understanding but was very limited in trying to account for the complex correlation inherent in longitudinal data. In contrast, the two-stage model allowed us to separate the different sources of variability to further understand individual patient responses and later population level responses. Finally, the linear mixed model could account for individual level variation, the time-based correlations observed as well as measurement error. This is therefore our recommendation for analysis of data of this nature. This final model indicated that age and iron deficiency had a negative significant effect on Hb trends over time, and Hb levels also significantly varied with past EPO doses (lagged dose), suggesting that EPO's impact fluctuates over time.

From a clinical perspective, these findings suggest that anemia management in hemodialysis patients should account for patient-specific factors such as age and iron status. Older patients may respond differently over time, and EPO treatment could be adjusted based on age-related Hb responses. The lack of a significant delayed effect of EPO dose indicates that immediate adjustments may be more effective than relying on past doses, supporting a focus on timely EPO adjustments based on current Hb levels. Given that iron-deficient patients show a stronger Hb response over time, regular monitoring and potential supplementation of iron are crucial for achieving target Hb levels. The study emphasizes the importance of individualized EPO and iron management strategies that adapt over time to optimize treatment outcomes for each patient.

A Appendix - Python/SAS Code

```
data hemo_data_lag;
    set hemo_data_poly;
    by ID;

    /* Initialize lagged variables to missing */
    retain lag_Hb lag_Dose;
    lag_Hb = .;
    lag_Dose = .;

    /* Calculate lag values only if not the first observation of each ID */
    if not first.ID then do;
        lag_Hb = lag(Hb);        /* Lag Hb from the previous observation */
        lag_Dose = lag(DOSE);    /* Lag Dose from the previous observation */
    end;

    /* Reset lag variables for the first observation of each ID */
    if first.ID then do;
```

```
        lag_Hb = .;
        lag_Dose = .;
    end;
run;
/* Summary statistics for continuous variables */
proc means data=hemo_data_lag n mean std min max nmiss;
    var Hb Age Lag_Hb Lag_Dose;
run;

/* 1. Mean Structure Analysis */
proc loess data=hemo_data_lag;
    ods output scoreresults=out;
    model Hb = MONTH; /* Using Hb as the response variable and MONTH as the predictor */
    score data=hemo_data_lag;
run;

proc sort data=out;
    by MONTH;
run;

proc sgplot data=out;
    series x=MONTH y=p_Hb / lineattrs=(color=black thickness=2) legendlabel="LOESS Smoothed
Hb";
    scatter x=MONTH y=Hb / markerattrs=(symbol=circlefilled color=red);
    xaxis label="Month";
    yaxis label="Hemoglobin (g/dl)";
    title "Hemoglobin Over Time with LOESS Smoothing";
run;

/* 2. Variance Structure Analysis */

proc loess data=hemo_data_lag;
    model Hb = MONTH AGE lag_Hb lag_Dose SEX IRON;
    output out=mean_residuals residual=residual;
run;

data mean_residuals;
    set mean_residuals;
    squared_residuals = residual**2;
    keep ID MONTH lag_Hb lag_Dose AGE SEX IRON residual squared_residuals;
run;


proc means data=mean_residuals noprint;
    class MONTH;
    var squared_residuals;
    output out=mean_variance mean=mean_squared_residuals;
run;
proc sgplot data=mean_variance;
    scatter x=MONTH y=mean_squared_residuals / markerattrs=(color=blue symbol=circlefilled);
    series x=MONTH y=mean_squared_residuals / lineattrs=(color=blue thickness=2);
    yaxis label="Mean Squared Residuals (Estimate of Variance)";
    xaxis label="Month";
    title "Unsmoothed Mean Variance Structure by Month";
```

```
run;

proc sgplot data=mean_variance;
    scatter x=MONTH y=mean_squared_residuals / markerattrs=(color=blue symbol=circlefilled)
legendlabel="Mean Squared Residuals";
    loess x=MONTH y=mean_squared_residuals / lineattrs=(color=black thickness=2) legendlabel="LOESS
Smoothed Variance";
    yaxis label="Mean Squared Residuals (Estimate of Variance)";
    xaxis label="Month";
    title "Smoothed Mean Variance Structure by Month";
run;

/* 3. Correlation Structure Analysis */
proc loess data=hemo_data_lag;
    model Hb = MONTH AGE lag_Hb lag_Dose SEX IRON;
    output out=mean_residuals residual=residual;
run;
data mean_residuals;
    set mean_residuals;
    keep ID MONTH residual;
run;
proc transpose data=mean_residuals out=residuals_wide prefix=Month_;
    by ID;
    id MONTH;
    var residual;
run;
proc sgscatter data=residuals_wide;
    matrix Month_1 Month_2 Month_3 Month_4 Month_5 Month_6 / diagonal=(histogram);
    title "Scatterplot Matrix of Residuals by Month";
run;

/*AUC*/
/* Step 1: Sort the dataset by ID and MONTH */
proc sort data=datasets.hemodialysis;
    by ID MONTH;
run;

/* Step 2: Calculate AUC for each individual using the trapezoidal rule */
data auc_with_covariates;
    set datasets.hemodialysis;
    by ID MONTH;

    /* Retain AUC to accumulate values for each ID */
    retain AUC 0;

    /* Initialize AUC for the first record of each ID */
    if first.ID then AUC = 0;

    /* Check for missing values in Hb */
    if missing(Hb) then do;
        output; /* Output the row, even if Hb is missing */
        return;
    end;
```

```
    /* Variables to hold the previous values */
    lag_Hb = lag(Hb); /* Previous Hb value */
    lag_MONTH = lag(MONTH); /* Previous month value */

    /* Calculate AUC using the trapezoidal rule, only if it's not the first month */
    if not first.ID and not missing(lag_Hb) and not missing(Hb) then do;
        AUC + (Hb + lag_Hb) / 2 * (MONTH - lag_MONTH);
    end;

    /* Output the record with AUC for the last observation of each ID */
    if last.ID then output;

    /* Keep relevant variables including all covariates */
    keep ID AUC SEX_dummy IRON Hb DOSE AGE MONTH;
run;

/* Step 3: Calculate total observation time for each individual */
proc sql;
    create table total_time as
    select ID,
           max(MONTH) - min(MONTH) as total_time format=8. /* Total time observed */
    from datasets.hemodialysis
    group by ID;
quit;

/* Step 4: Merge AUC with total time and calculate standardized AUC */
data final_auc;
    merge auc_with_covariates total_time;
    by ID;

    /* Calculate standardized AUC */
    if total_time > 0 then standardized_auc = AUC / total_time;
    else standardized_auc = .; /* Handle cases with no time interval */

    /* Keep relevant variables for analysis */
    keep ID standardized_auc AUC SEX_dummy IRON Hb AGE DOSE MONTH total_time;
run;

/* Step 5: Multiple Linear Regression with Adjustment for Covariates using PROC GLM for Standardized
AUC */
proc glm data=final_auc;
    class SEX_dummy IRON;  /* Ensure SEX_dummy is treated as a categorical variable */
    model standardized_auc = DOSE SEX_dummy IRON AGE / solution;  /* Adjust for covariates
*/
    title "Multiple Linear Regression of Standardized AUC with Adjustment for Covariates";
run;

/*INCREMENTS*/
/* Step 1: Calculate the Hb Increment for Each Patient */
data increments;
    set datasets.hemodialysis;
    by ID;

    /* Retain only the first and last Hb values for each patient */
```

```
    retain first_Hb last_Hb;

    /* Initialize first Hb for the first observation of each patient */
    if first.ID then first_Hb = Hb;

    /* Capture the last Hb value for each patient */
    if last.ID then do;
        last_Hb = Hb;
        /* Calculate the increment (change in Hb from first to last observation) */
        Hb_increment = last_Hb - first_Hb;
        output; /* Output the increment record for each patient */
    end;

    /* Keep relevant variables for analysis */
    keep ID Hb_increment SEX_dummy AGE IRON DOSE;
run;

/* Step 2: Multiple Linear Regression of Hb Increment with Covariates */
proc glm data=increments;
    class SEX_dummy IRON;  /* Ensure SEX_dummy is treated as a categorical variable */
    model Hb_increment = SEX_dummy AGE IRON DOSE / solution;
    title "Multiple Linear Regression of Hb Increment (First to Last Observation) with Covariates";
run;

/*multivariate initial model*/

/* MODEL 1 SATURATED */
proc mixed data=datasets.hemodialysis_with_lag method=ml;
    class ID month_copy SEX_dummy IRON;
    model Hb = month month*month AGE SEX_dummy IRON lag_hb lag_dose
               month*AGE month*SEX_dummy month*lag_DOSE month*IRON month*lag_hb
               month*month*AGE month*month*SEX_dummy month*month*lag_DOSE month*month*IRON
month*month*lag_hb/ solution;
    repeated month_copy / type=un subject=ID;
run;

/*model 2- NO month*month*AGE interaction*/
proc mixed data=datasets.hemodialysis_with_lag method=ml;
    class ID month_copy SEX_dummy IRON;
    model Hb = month month*month AGE SEX_dummy IRON lag_hb lag_dose
               month*AGE month*SEX_dummy month*lag_DOSE month*IRON month*lag_hb
               month*month*SEX_dummy month*month*lag_DOSE month*month*IRON MONTH*MONTH*LAG_HB/
solution;
    repeated month_copy / type=un subject=ID;
run;

/* EQUAL LRT SO REMOVE MONTH*MONTH*AGE interaction */

/*model 3 - NO month*month*lag_DOSE interaction*/
proc mixed data=datasets.hemodialysis_with_lag method=ml;
    class ID month_copy SEX_dummy IRON;
    model Hb = month month*month AGE SEX_dummy IRON lag_hb lag_dose
               month*AGE month*SEX_dummy month*lag_DOSE month*IRON month*lag_hb
               month*month*SEX_dummy month*month*IRON MONTH*MONTH*LAG_HB / solution;
```

```
    repeated month_copy / type=un subject=ID;
run;

/**REMOVE MONTH*MONTH*LAG_DOSE**/

/*model 4 - NO month*month*SEX interaction*/
proc mixed data=datasets.hemodialysis_with_lag method=ml;
    class ID month_copy SEX_dummy IRON;
    model Hb = month month*month AGE SEX_dummy IRON lag_hb lag_dose
               month*AGE month*SEX_dummy month*lag_DOSE month*IRON month*lag_hb
               month*month*IRON MONTH*MONTH*LAG_HB / solution;
    repeated month_copy / type=un subject=ID;
run;

/*REMOVE MONTH*MONTH*SEX */

/*MODEL 5 - NO MONTH*MONTH*IRON */
proc mixed data=datasets.hemodialysis_with_lag method=ml;
    class ID month_copy SEX_dummy IRON;
    model Hb = month month*month AGE SEX_dummy IRON lag_hb lag_dose
               month*AGE month*SEX_dummy month*lag_DOSE month*IRON month*lag_hb
               month*month*LAG_HB/ solution;
    repeated month_copy / type=un subject=ID;
run;

/* REMOVE MONTH*MONTH*IRON */

/* MODEL 6 NO MONTH*MONTH*LAG_HB */
proc mixed data=datasets.hemodialysis_with_lag method=ml;
    class ID month_copy SEX_dummy IRON;
    model Hb = month month*month AGE SEX_dummy IRON lag_hb lag_dose
               month*AGE month*SEX_dummy month*lag_DOSE month*IRON month*lag_hb
               /solution;
    repeated month_copy / type=un subject=ID;
run;

/* KEEP MONTH*MONTH*LAG_HB */
/*MODEL 5*/
proc mixed data=datasets.hemodialysis_with_lag method=ml;
    class ID month_copy SEX_dummy IRON;
    model Hb = month month*month AGE SEX_dummy IRON lag_hb lag_dose
               month*AGE month*SEX_dummy month*lag_DOSE month*IRON month*lag_hb
               month*month*LAG_HB/ solution;
    repeated month_copy / type=un subject=ID;
run;

/* model 7 - NO month*month interaction */
proc mixed data=datasets.hemodialysis_with_lag method=ml;
    class ID month_copy SEX_dummy IRON;
    model Hb = month AGE SEX_dummy IRON lag_hb lag_dose
               month*AGE month*lag_DOSE month*IRON month*lag_hb
               month*month*LAG_HB / solution;
    repeated month_copy / type=un subject=ID;
run;
```

```
/* KEEP MONTH*MONTH  */
/*MODEL 5*/
proc mixed data=datasets.hemodialysis_with_lag method=ml;
    class ID month_copy SEX_dummy IRON;
    model Hb = month month*month AGE SEX_dummy IRON lag_hb lag_dose
               month*AGE month*SEX_dummy month*lag_DOSE month*IRON month*lag_hb
               month*month*LAG_HB/ solution;
    repeated month_copy / type=un subject=ID;
run;
/*

/*MODEL 8 NO MONTH*AGE -*/
proc mixed data=datasets.hemodialysis_with_lag method=ml;
    class ID month_copy SEX_dummy IRON;
    model Hb = month month*month AGE SEX_dummy IRON lag_hb lag_dose
               month*SEX_dummy month*lag_DOSE month*IRON month*lag_hb
               month*month*LAG_HB/ solution;
    repeated month_copy / type=un subject=ID;
run;

/*REMOVE MONTH*AGE */

/* MODEL 9 NO MONTH*SEX_DUMMY */
proc mixed data=datasets.hemodialysis_with_lag method=ml;
    class ID month_copy SEX_dummy IRON;
    model Hb = month month*month AGE SEX_dummy IRON lag_hb lag_dose
               month*lag_DOSE month*IRON month*lag_hb
               month*month*LAG_HB/ solution;
    repeated month_copy / type=un subject=ID;
run;

/* REMOVE MONTH*SEX_dummy */

/* MODEL 10 NO MONTH*IRON */
proc mixed data=datasets.hemodialysis_with_lag method=ml;
    class ID month_copy SEX_dummy IRON;
    model Hb = month month*month AGE SEX_dummy IRON lag_hb lag_dose
               month*lag_DOSE month*lag_hb
               month*month*LAG_HB/ solution;
    repeated month_copy / type=un subject=ID;
run;


/* REMOVE MONTH*IRON */

/*MODEL 11 SECOND TRY NO MONTH*MONTH*/
proc mixed data=datasets.hemodialysis_with_lag method=ml;
    class ID month_copy SEX_dummy IRON;
    model Hb = month AGE SEX_dummy IRON lag_hb lag_dose
               month*lag_DOSE month*lag_hb
               month*month*LAG_HB/ solution;
    repeated month_copy / type=un subject=ID;
run;
```

```
/*KEEP MONTH*MONTH*/
proc mixed data=datasets.hemodialysis_with_lag method=ml;
    class ID month_copy SEX_dummy IRON;
    model Hb = month month*month AGE SEX_dummy IRON lag_hb lag_dose
               month*lag_DOSE month*lag_hb
               month*month*LAG_HB/ solution;
    repeated month_copy / type=un subject=ID;
run;

/*REDUCE COVARIANCE STR FROM MODEL 10*/

/* model 12 - ARH(1) covariance */
proc mixed data=datasets.hemodialysis_with_lag method=ml;
    class ID month_copy SEX_dummy IRON;
    model Hb = month month*month AGE SEX_dummy IRON lag_hb lag_dose
               month*lag_DOSE month*lag_hb
               month*month*LAG_HB/ solution;
    repeated month_copy / type=ARH(1) subject=ID;
run;

/* model 13 - Toeplitz H covariance */
proc mixed data=datasets.hemodialysis_with_lag method=ml;
    class ID month_copy SEX_dummy IRON;
    model Hb = month month*month AGE SEX_dummy IRON lag_hb lag_dose
               month*lag_DOSE month*lag_hb
               month*month*LAG_HB/ solution;
    repeated month_copy / type=TOEPH subject=ID;
run;

/*REML*/
/* model 14'' - TOEPH Covariance*/
proc mixed data=datasets.hemodialysis_with_lag method=reml;
    class ID month_copy SEX_dummy IRON;
    model Hb = month month*month AGE SEX_dummy IRON lag_hb lag_dose
               month*lag_DOSE month*lag_hb
               month*month*LAG_HB/ solution;
    repeated month_copy / type=TOEPH subject=ID;
run;


/*multivariate final model*/

/*FINAL MODEL : MODEL 13*/
/* Run without empirical to get normal standard errors */
proc mixed data=datasets.hemodialysis_with_lag method=ml covtest;
    class ID month_copy SEX_DUMMY(ref = "0") IRON (ref = "0");
    model Hb = month AGE SEX_dummy IRON lag_hb lag_dose
               month*month month*lag_dose month*lag_hb month*month*lag_hb/ solution;
    repeated month_copy / type=TOEPH subject=ID;
    ods output solutionf=sf_normal;  /* Save standard errors without empirical */
run;
```

```
/* Run with empirical to get robust standard errors */
proc mixed data=datasets.hemodialysis_with_lag method=ml covtest empirical;
    class ID month_copy SEX_dummy(ref = "0") IRON(ref = "0");
    model Hb = month AGE SEX_dummy IRON lag_hb lag_dose
               month*month month*lag_dose month*lag_hb month*month*lag_hb/ solution;
    repeated month_copy / type=TOEPH subject=ID;
/*     ods output solutionf=sf_robust;  /* Save standard errors with empirical */
run;



/*two-stage*/


proc sort data=mydata;
    by ID MONTH;
run;


data mydata_lagged;
    set mydata;
    by ID;

    retain Prev_Hb Prev_Dose;

    /* Initialize Lagged_Hb and Lagged_Dose */
    Lagged_Hb = Prev_Hb;
    Lagged_Dose = Prev_Dose;

    /* Update Prev_Hb and Prev_Dose with current values */
    Prev_Hb = Hb;
    Prev_Dose = DOSE;

    if first.ID then do;
        Lagged_Hb = .;
        Lagged_Dose = .;
        Prev_Hb = Hb;
        Prev_Dose = DOSE;
    end;

    /* If MONTH = 1, set lagged variables to missing */
    if MONTH = 1 then do;
        Lagged_Hb = .;
        Lagged_Dose = .;
    end;
run;


if necessary */
data mydata_stage1;
    set mydata_lagged;
    if MONTH > 1;
run;
```

```
proc reg data=mydata_stage1 noprint outest=Estimates rsquare adjrsq aic bic;
    by ID;
    model Hb = MONTH IRON Lagged_Dose lagged_hb ;
    output out=Predictions predicted=PredictedHb residual=Residuals;
run;

proc reg data=mydata_stage1;
    model Hb = MONTH IRON Lagged_Dose Lagged_Hb / vif;
run;


data Stage1_Coefficients;
    set Estimates;
    keep ID Intercept MONTH IRON lagged_DOSE lagged_hb;
    rename MONTH=Slope_Month IRON=Coef_Iron lagged_DOSE=Coef_Dose lagged_hb=Coef_hb;
run;

proc means data=Stage1_coefficients;
    var intercept Slope_Month Coef_Iron Coef_Dose Coef_hb;
run;


proc sort data=mydata nodupkey out=Subject_Covariates(keep=ID AGE SEX);
    by ID;
run;

data Stage2_Data;
    merge Stage1_Coefficients Subject_Covariates;
    by ID;
run;





/*linear mixed*/
/* Tests for Serial Correlation */

/* Model 1: Measurement Error Only (No Serial Correlation) */
proc mixed data=hemo_data_lag method=reml;
    class ID month_class SEX IRON;
    model Hb = month AGE SEX IRON lag_Dose lag_Hb
               month*AGE
               month*lag_Dose
               month*lag_Hb
               month*IRON  / solution;
    random intercept month / type=un subject=ID g gcorr v vcorr;
    repeated month_class / local subject=ID r rcorr;
run;

/* Model 2: Measurement Error + Exponential Serial Correlation */
proc mixed data=hemo_data_lag method=reml;
    class ID month_class SEX IRON;
    model Hb = month  AGE SEX IRON lag_Dose lag_Hb
```

```
               month*AGE
               month*lag_Dose
               month*lag_Hb
               month*IRON  / solution;
    random intercept month / type=un subject=ID g gcorr v vcorr;
    repeated month_class / type=sp(exp)(month) local subject=ID r rcorr;
run;

/* Model 3: Measurement Error + Gaussian Serial Correlation */
proc mixed data=hemo_data_lag method=reml;
    class ID month_class SEX IRON;
    model Hb = month  AGE SEX IRON lag_Dose lag_Hb
               month*AGE
               month*lag_Dose
               month*lag_Hb
               month*IRON / solution;
    random intercept month / type=un subject=ID g gcorr v vcorr;
    repeated month_class / type=sp(gau)(month) local subject=ID r rcorr;
run;

/* Test for Random effects*/

/* Model 1: Random Intercept Only */
proc mixed data=hemo_data_lag method=reml;
    class ID month_class SEX IRON;
    model Hb = month AGE SEX IRON lag_Dose lag_Hb
               month*AGE month*lag_Dose month*IRON month*lag_Hb / solution;
    random intercept / subject=ID g gcorr v vcorr;
    repeated month_class / type=sp(gau)(month) local subject=ID r rcorr;
run;

/* Model 2: Random Intercept + Random Slopes */
proc mixed data=hemo_data_lag method=reml;
    class ID month_class SEX IRON;
    model Hb = month AGE SEX IRON lag_Dose lag_Hb
               month*AGE month*lag_Dose month*IRON month*lag_Hb / solution;
    random intercept month  / type=un subject=ID g gcorr v vcorr;
    repeated month_class / type=sp(gau)(month) local subject=ID r rcorr;
run;

/*Reduction of mean structure*/
proc mixed data=hemo_data_lag method=reml;
    class ID month_class SEX IRON;
    model Hb = month month2 AGE SEX IRON lag_Dose lag_Hb
               month_age month_sex month_iron month_lag_dose month_lag_hb
               month2_age month2_sex month2_iron month2_lag_dose month2_lag_hb / solution;
    random intercept / subject=ID type=un g gcorr v vcorr;
    repeated month_class / type=sp(gau)(month) local subject=ID r rcorr;
run;
/*FINAL MODEL*/
proc mixed data=hemo_data_lag method=reml;
    class ID month_class SEX(ref="0") IRON(ref="0");
    model Hb = month month2 AGE SEX IRON lag_dose lag_hb
               month_age month_iron month2_iron month_lag_dose / solution;
```

```
    random intercept / subject=ID type=un g gcorr v vcorr;
    repeated month_class / type=sp(gau)(month) local subject=ID r rcorr;
run;
```